HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **horizontal scaling** strategies for distributed in-memory data structures, improving throughput by **42%** under peak load.
- Developed **data sharding** algorithms to distribute datasets across nodes, ensuring consistent low-latency access for over 10,000 concurrent users.
- Implemented **robust authentication and authorization protocols** using secure token-based mechanisms to protect sensitive data within the managed service.
- Contributed to the open-source **Valkey** community by developing and testing net-new features for core data replication functionalities.
- Optimized **memory management** and garbage collection within the in-memory database engine, reducing memory footprint by **18%**.
- Wrote production runbooks and internal documentation that the customer onboarding team now uses to bring new clients live in ~10 days instead of 4+ weeks.
- Containerised the feature-pipeline services with **Docker** and deployed them to a **Kubernetes** cluster (EKS), wiring up Helm chart configuration and basic resource limits for predictable rollout.

SKILLS

Core Technologies – Valkey, Redis OSS, Memcached, Java, Python, SQL, AWS ElastiCache, AWS MemoryDB

Programming Languages – Java, Python, SQL

Cloud & DevOps – AWS, Docker, Kubernetes, Jenkins, Git, Linux

Databases & Data Stores – PostgreSQL, MySQL, Oracle, In-memory databases

Concepts – Distributed Computing, Data Storage, High Availability, Data Replication, Data Sharding, Authentication, Authorization, Memory Management, Horizontal Scaling, Vertical Scaling

PROJECTS

Distributed Caching Layer for High-Throughput Applications

- Engineered a distributed caching layer using **Valkey OSS** (compatible with ElastiCache) to serve frequently accessed data, reducing database load by **60%**.
- Implemented advanced caching strategies including cache invalidation and TTL management to ensure data consistency across a cluster of 15 nodes.
- Developed **REST APIs in Java Spring Boot** to interface with the caching layer, handling approximately 5,000 requests per second.

Performance Optimization of a Key-Value Store

- Analyzed performance bottlenecks in a **Redis OSS**-based key-value store, identifying suboptimal data retrieval patterns.
- Refactored data access code and optimized **SQL queries** used for populating cache, leading to a **35%** reduction in data refresh times.
- Contributed to the development of monitoring dashboards using **Splunk** to track cache hit rates and latency, ensuring operational visibility.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)